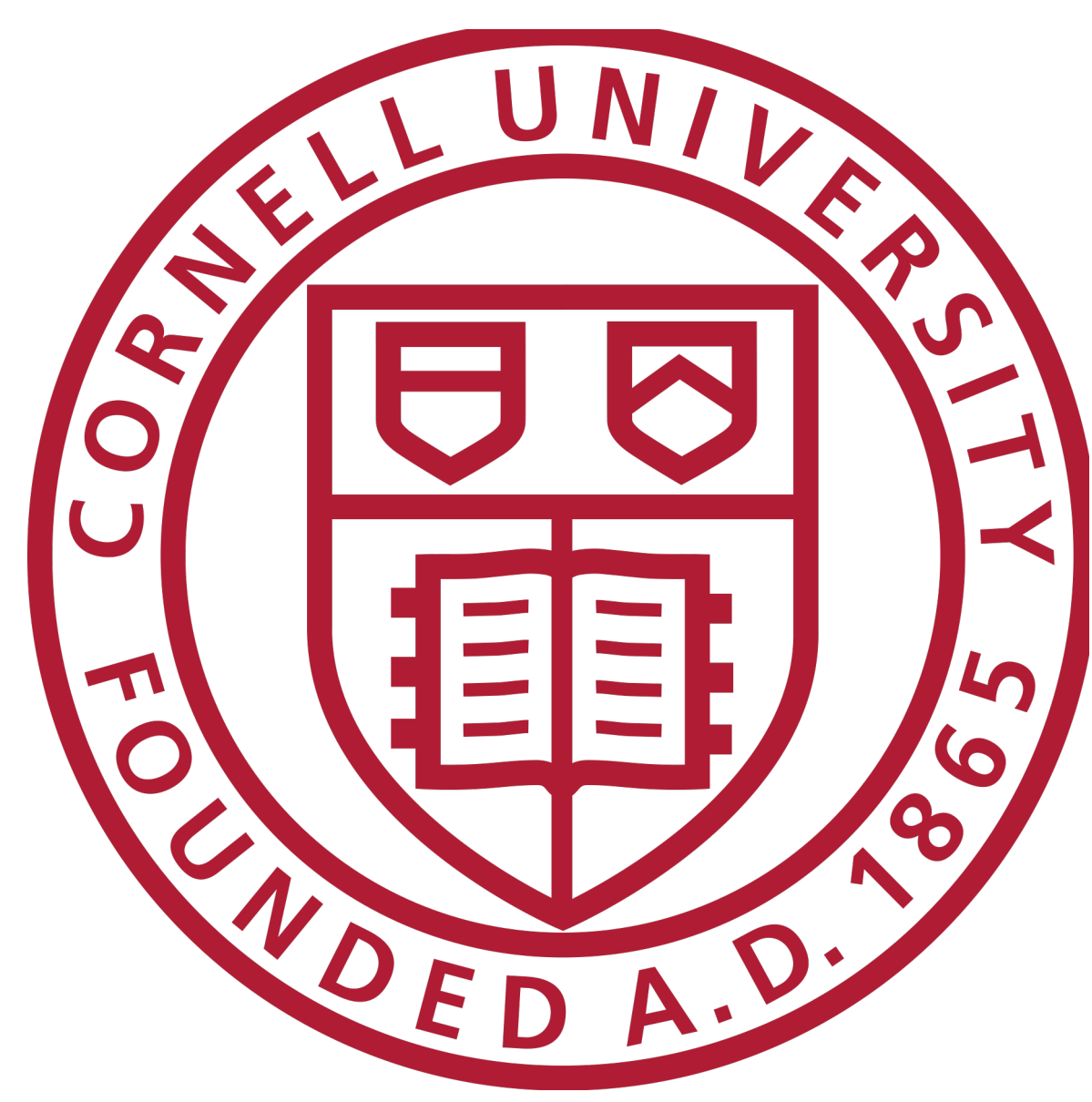
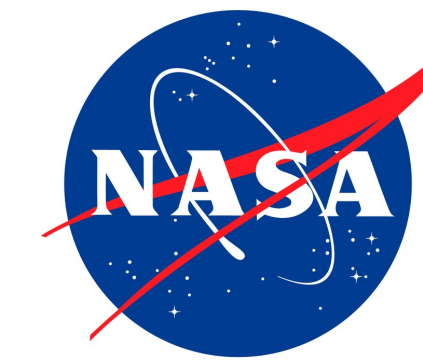




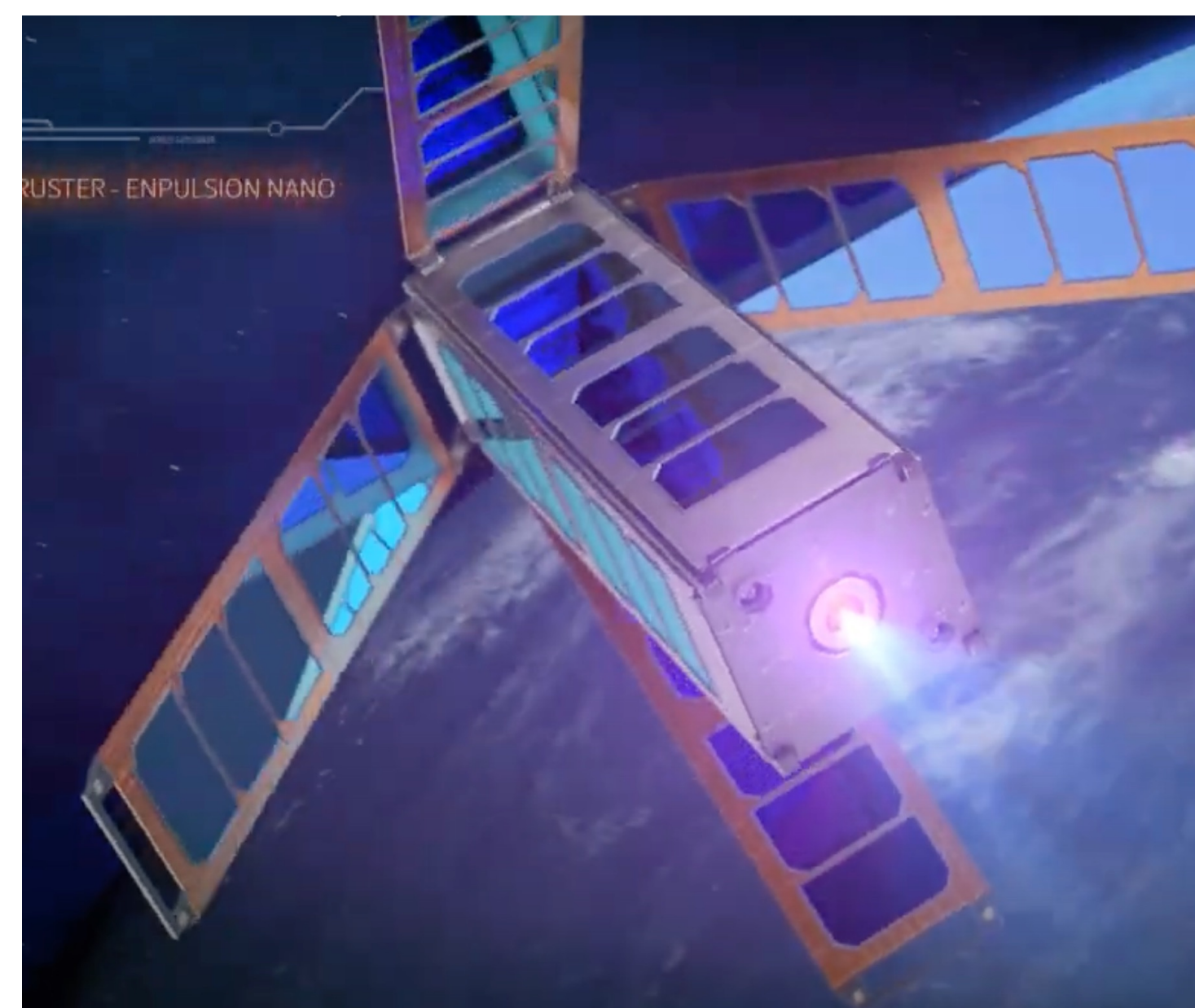
Quantifying Electrospray Plume–Surface Interactions Through Secondary Target Analysis

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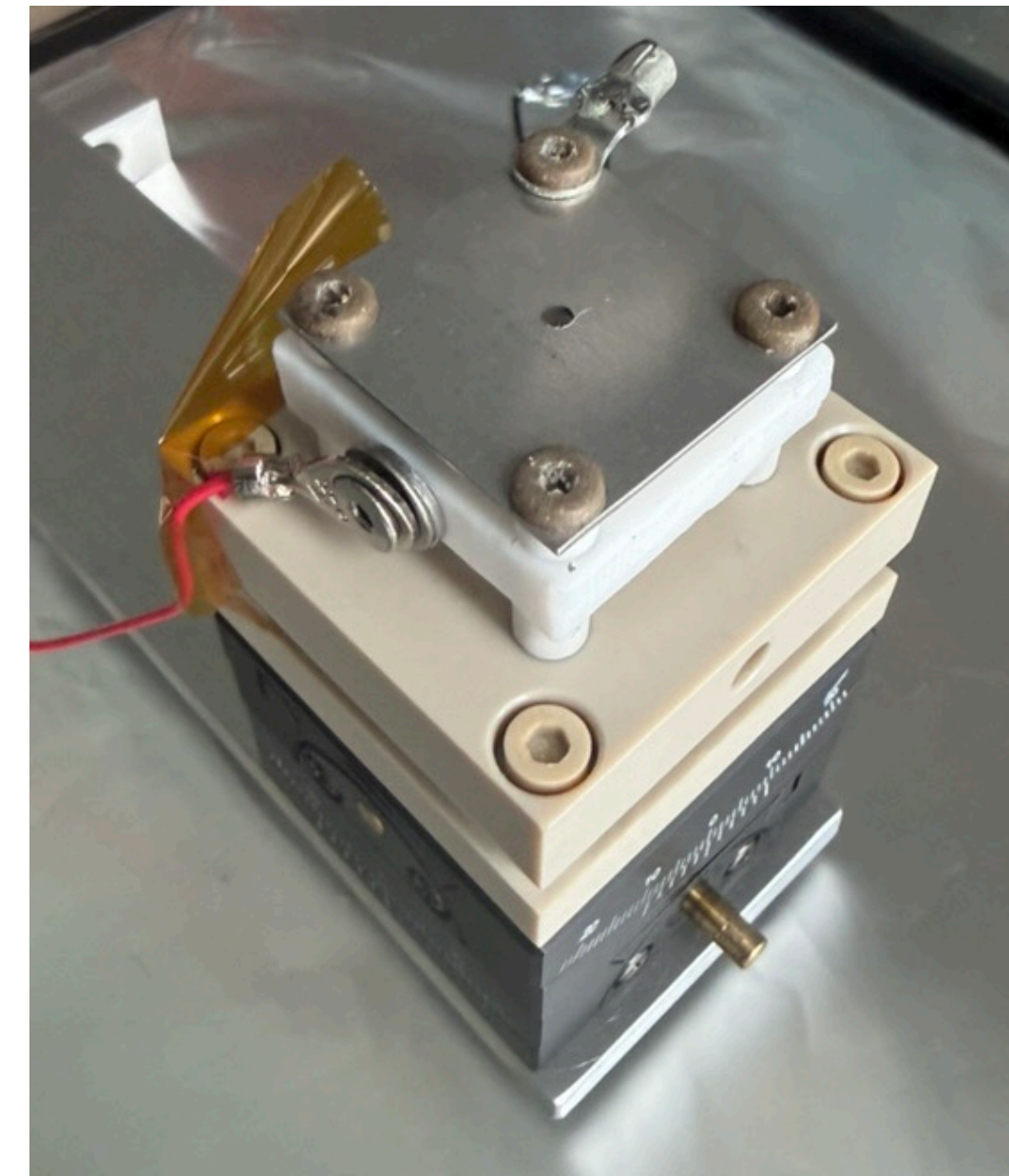


Introduction

Electrospray propulsion is an electrostatic space propulsion technology that generates thrust by accelerating charged ions or droplet from a liquid propellant through sharp micro-scale emitters¹. While electrosprays are well suited for applications like small satellite propulsion and precise attitude adjustment, plume-surface interactions – either with reactive spacecraft surfaces or ground testing devices– are not fully characterized².



Concept of an Enpulsion Field Emission Electrospray Thruster.
Image Credit: Enpulsion

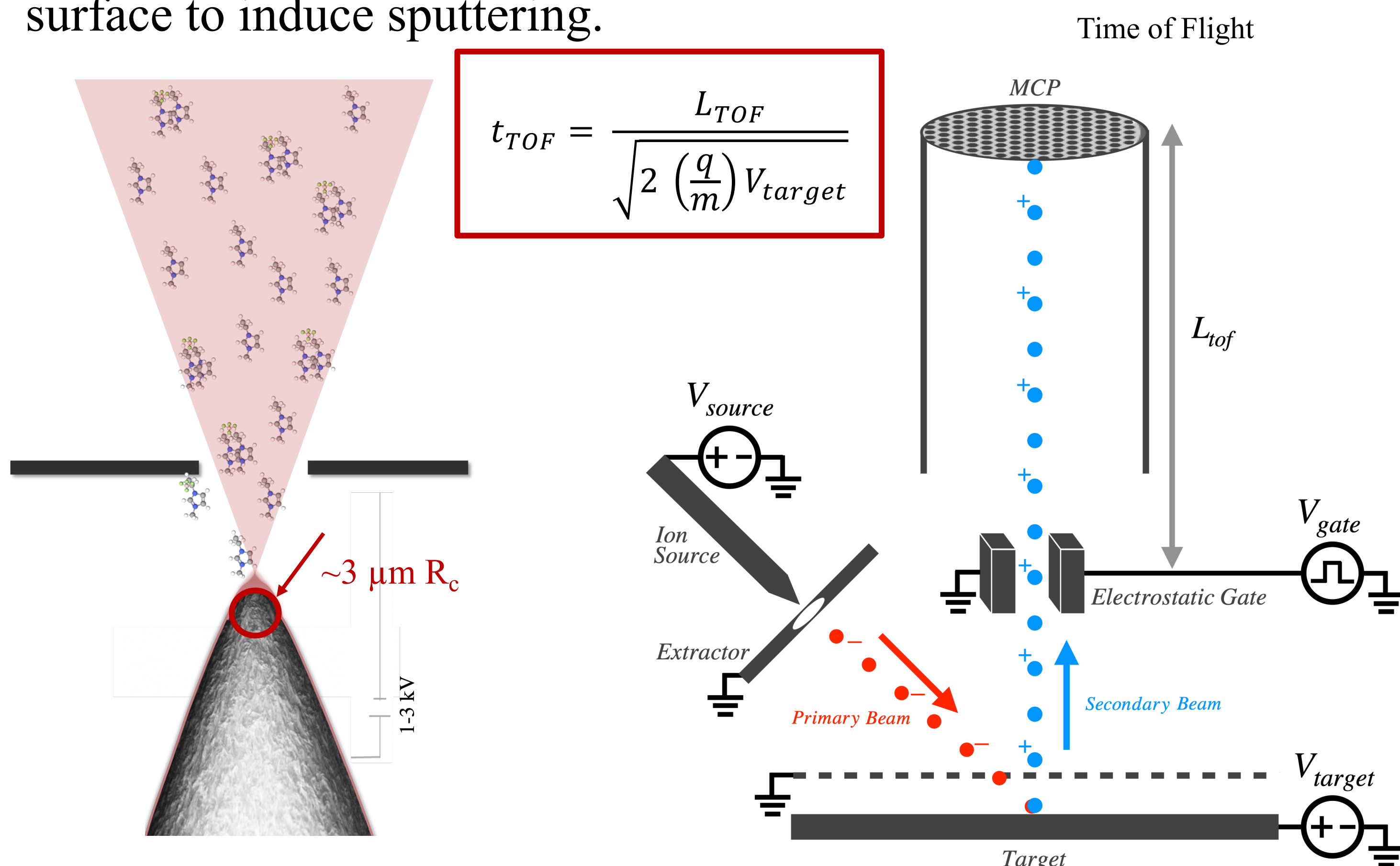


Single Emitter Source Setup

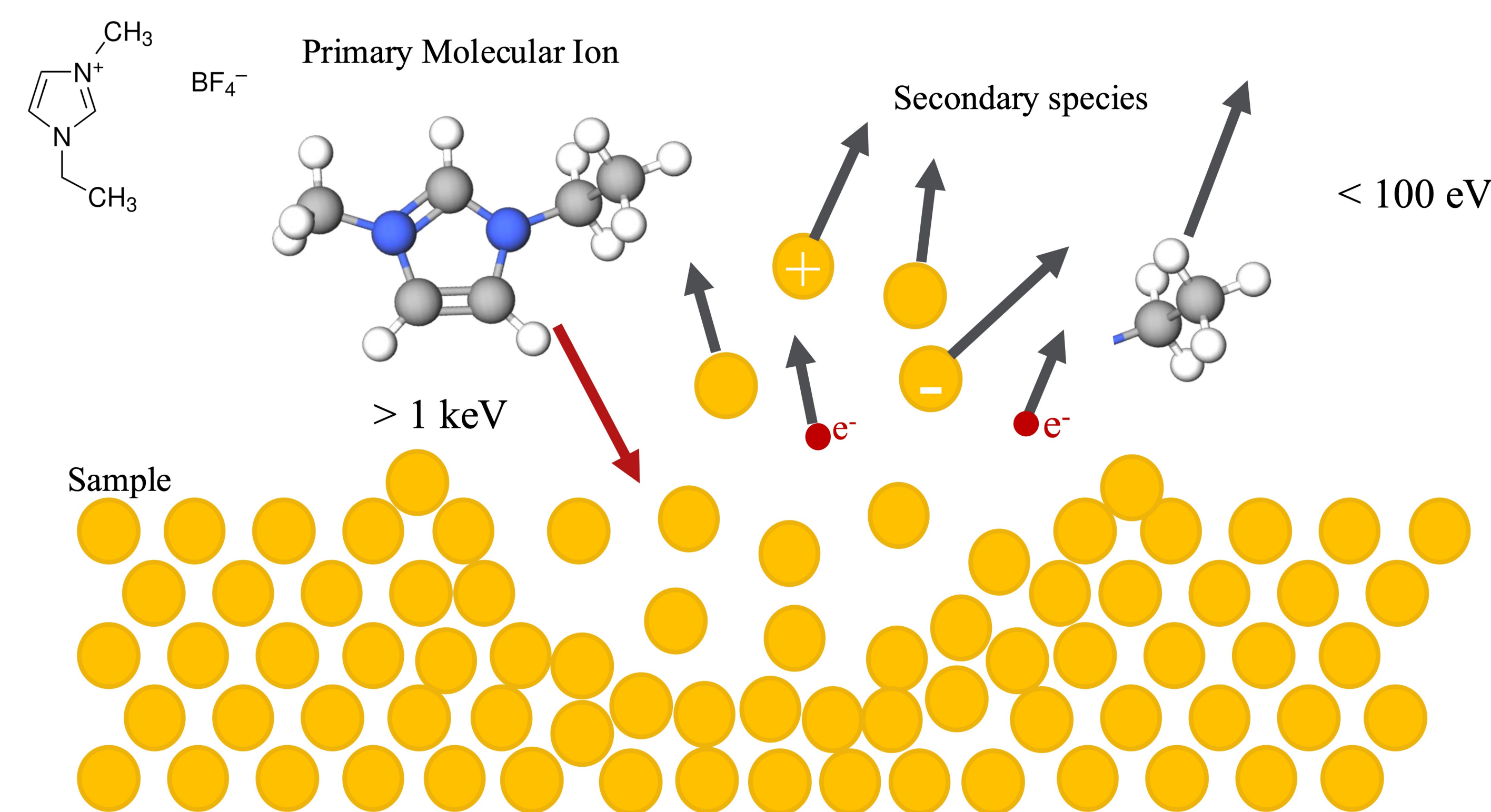
Methodology

Electrospray Time-of-Flight Secondary Ion Mass Spectrometry (ESI TOF-SIMS) was used to investigate electrospray plume-surface interactions, where a high-energy primary ion beam impacts a surface, inducing particle sputtering.⁴ For ESI TOF-SIMS, the primary source consists of a room temperature ionic liquid ion source, creating a polydisperse molecular ion plume. This high energy (>1 keV) plume is directed at a material surface to induce sputtering.

$$t_{TOF} = \frac{L_{TOF}}{\sqrt{2 \left(\frac{q}{m} \right) V_{target}}}$$

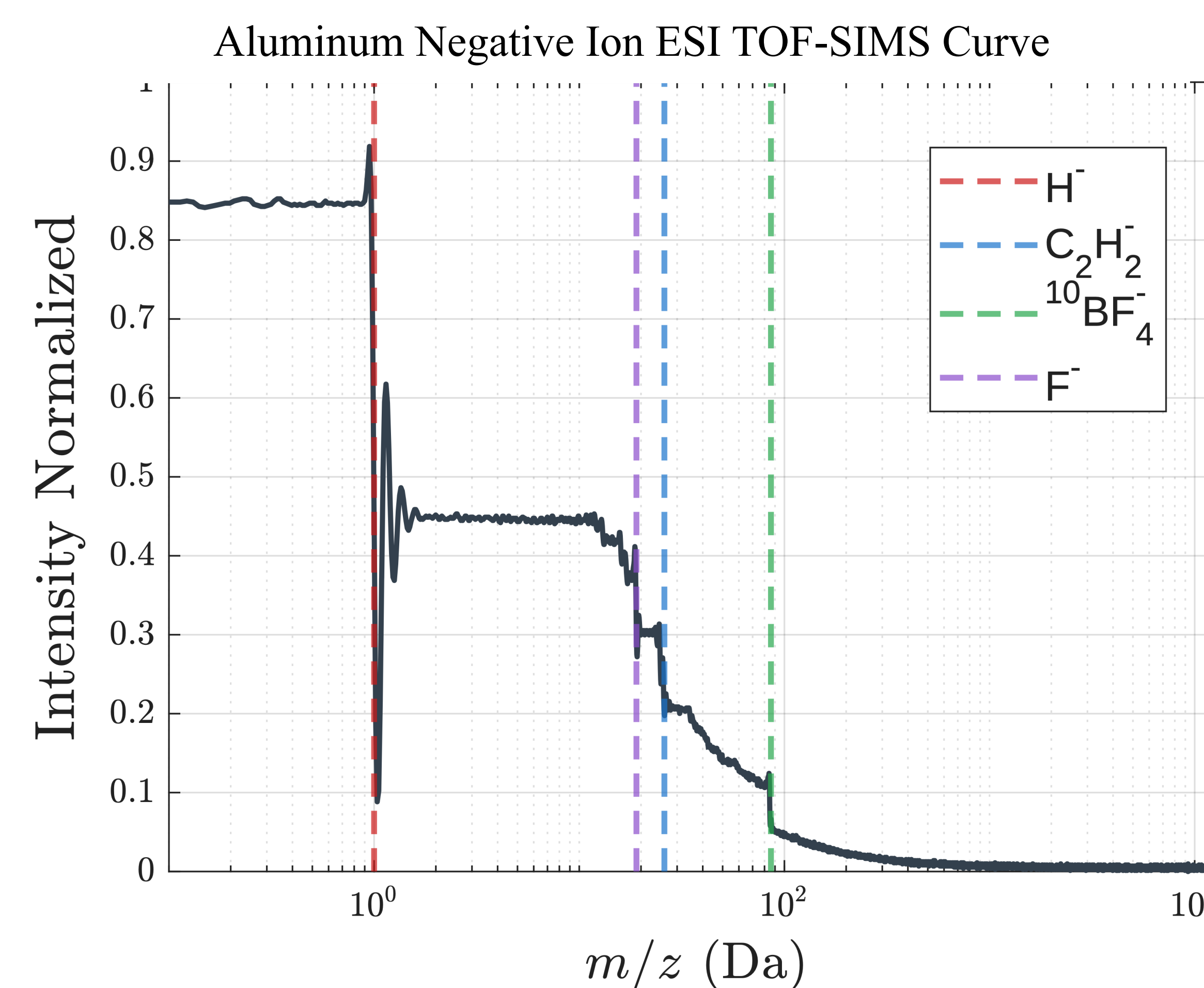


Methodology



A portion of the sputtered species exist as secondary ions of both polarities. These ions are extracted via an acceleration grid and directed to an in-house TOF mass spectrometer with a microchannel plate detector. The mass of the secondary ion species are then derived from conservation of energy principles.

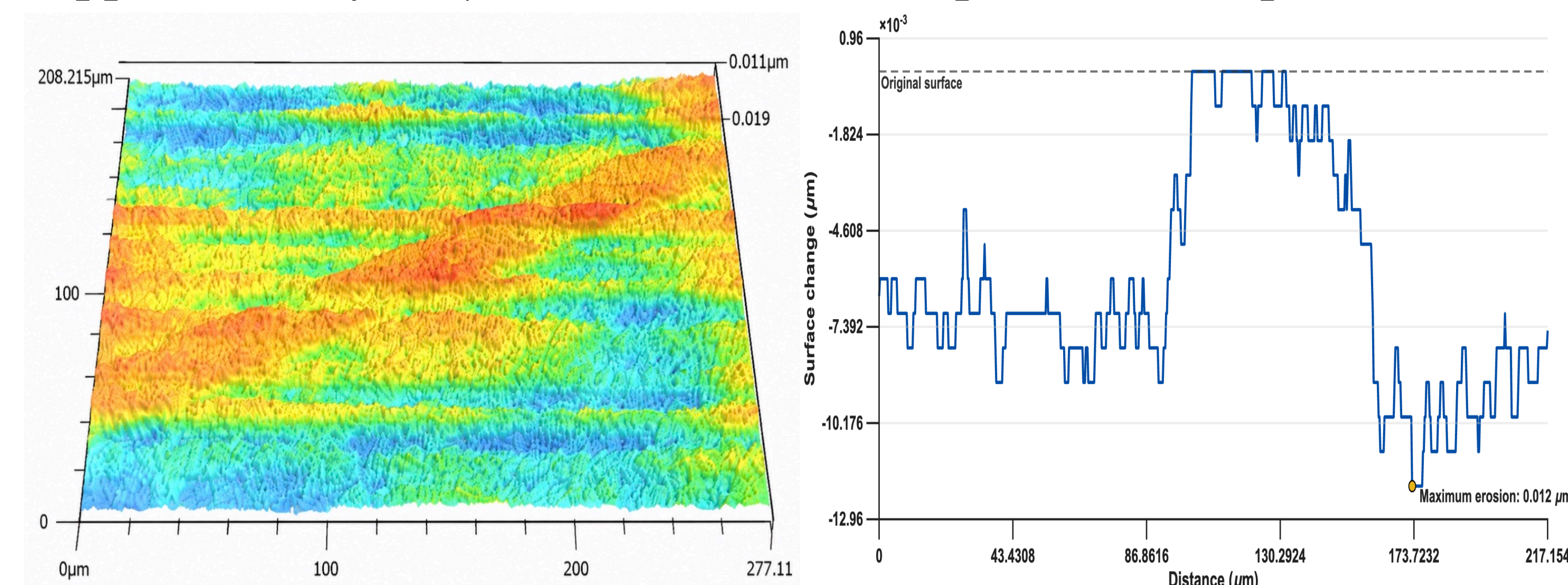
Results



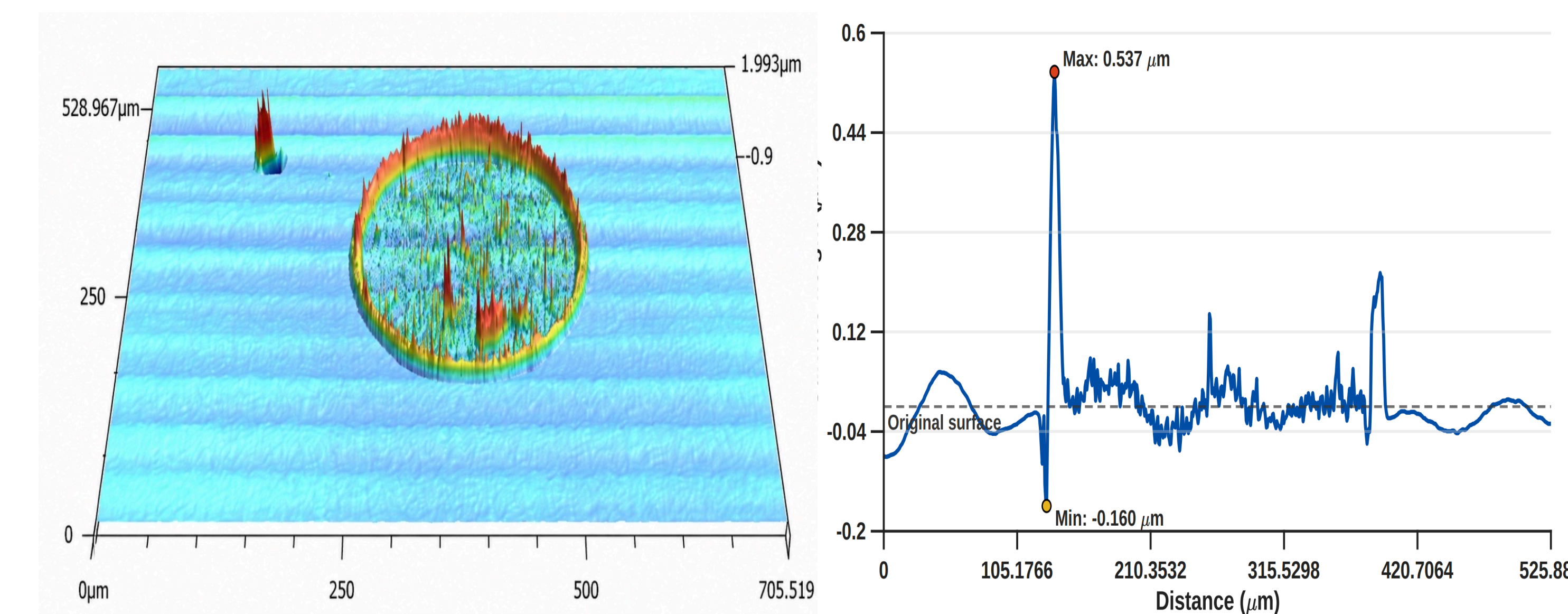
Preliminary analysis shows secondary ions below 100 amu for negative polarity for an ionic liquid electrospray plume impacting an aluminum target. The mass spectral ‘families’ in the negative polarity mode indicate propellant ions, with mass lines indicating the potential presence intact anion, fluorine, and hydrocarbon fragments.

Results

Preliminary optical profilometry of tungsten samples directly exposed to the electrospray plume revealed consistent surface erosion within the plume impact region. Comparison with unexposed control regions showed erosion depths of approximately 10 μm for 16 hours of exposure to the plume.



Optical profilometry revealed localized ionic liquid droplet deposition on both tungsten and aluminum surfaces. Deposits on the aluminum wafer reached heights of up to 0.537 μm , demonstrating material accumulation alongside erosion.



Conclusion

Electrospray propulsion devices show promise to be mission-enabling technologies. Characterizing plume-surface interactions will improve understanding of their effects on spacecraft materials and help quantify propellant source lifetimes.

References & Acknowledgements

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